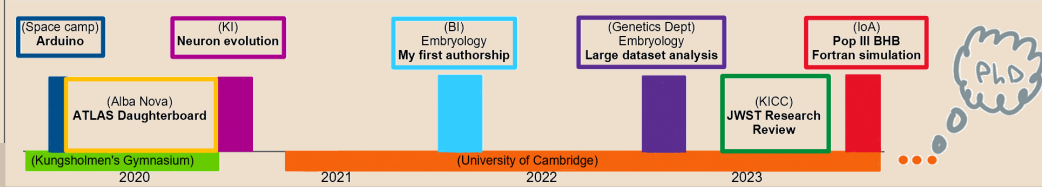


Sebastian Ljung

Natural Sciences at Cambridge

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Education

University of Cambridge, Peterhouse

- Natural Sciences, doing 4th year integrated masters in Astrophysics. ~11 months internship experience in embryology (statistical analysis and clustering) and astrophysics.
- **Research review** (1st class), Kavli Insitute for Cosmology [4 months, part-time]: 2022
 - ↳ On first galaxies with JWST, with [Dr. Laporte](#) (Prof. Tacchella's group). >28 months of group meetings (still going)
- **Computing project**^(AI) (1st class) [4 weeks]: Vorticies in Incompressible Schrödinger Flow. 2024
 - ↳ Implemented model from a paper to investigate interplay between vortex element evolution & plank length. Used $\mathbb{C}^2$ field solutions to a modified Schrödinger eq. as hydrodynamic simulation.
- Masters modules (Math dept): Advanced Stellar Evolution, Astro-statistics, Rel. astrophysics & cosmology +lectures on: GR, & QFT.

Publications

- Taubenschmid-Stowers J., Rostovskaya M., Santos F., Ljung S., ..., Reik Wolf 2022
"Modelling human zygotic genome activation in 8C-like cells in vitro"

Research Experience

- **MSc project supervised by Prof. Tacchella & Dr. Gilkis, Kavli Institute of Cosmology:**
 - ↳ Part of collaboration, (modifying MESA stars code, in Fortran):
 - (For Stars group): Implement star-disk model to enable stable modelling of differentially rotating binaries.
 - (For Extragalactic group): Then generate grid of binaries, used for spectral fitting of high-z galaxies.
 - Independently formulated analytic solution to Prof. Paczynski's (1991) numerical model on polytropic accretion disks around critically rotating main sequence (MS) stars.
 - Incorporated supernovae & evolution of compact object binaries within MESA.
- **Internship in Prof. Anastasia Fialkov's group, Institute of Astronomy [8 weeks]:** 2023
 - ↳ Modified semi-analytic A-SLOTH simulation (Fortran) to account for the influence of super-massive black holes (SMBH) on the modelled nuclear stellar clusters & Pop III black hole binaries.
 - After literature review, proposed a side-project to introduce SMBHs
 - Incorporated equations from publications and justified modelling approximations.
 - Implemented and argued for significance of eccentricity on Stochastic GW Background.
 - Prof. Roman Rafikov recently (Oct) offered to help guide this to publication, after MSc.
- **Internship at Dr Michael Imbeault lab, Genetics Dept. Cambridge [10 weeks]:** 2022
 - Handling large datasets (>30GB), w/o cluster, in a RAM and time-efficient manner.
 - Spearheaded side-project, and liaised with Prof. Reik's, lab, to determine potential ZGA targets for collaboration, using developed tool.
- **Internship at Prof. Wolf Reik lab, BI Cambridge [9 weeks]:** 2021
 - ↳ Suggested & applied tool that uses 1st-order ODEs & dimensionality reduction to predict cell state transitions. (analogous to phase-space flow in galactic dynamics.)
 - Co-authorship for answering review question, computationally bypassing months of costly experiments.
 - Presented results of experimental + computational (side-project) at conference poster session.